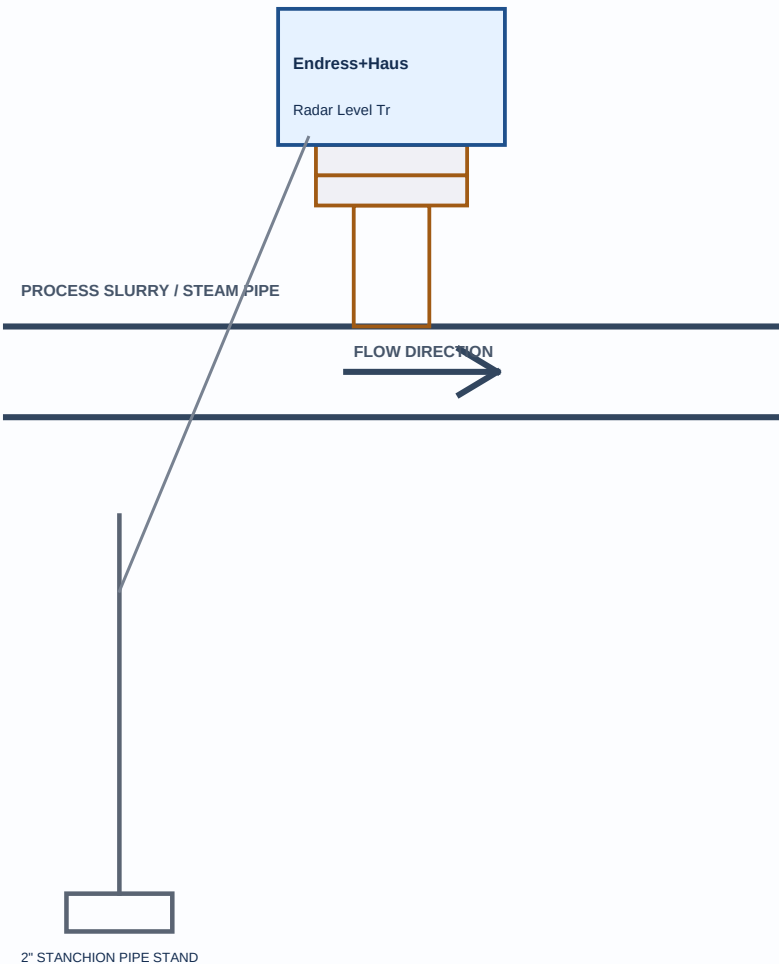


TEMPLATE TYP-14: RADAR LEVEL TRANSMITTER

TYPICAL TAGS: LT-40101 / LT-40201

SERVICE: Cooker Slurry & Flash Tank Non-Contact Continuous Level | PLC I/O: 1756-IF16 (16-Point Differential AI)

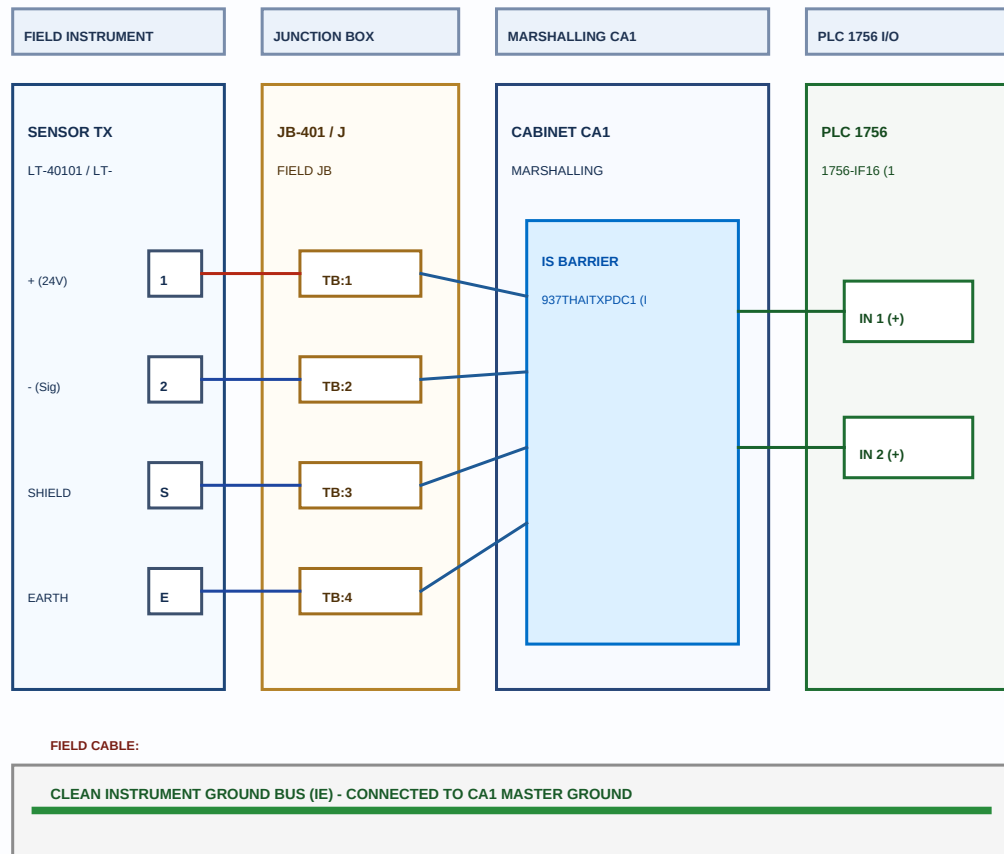
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	DN80 (3") ASME Cl.150 RF Fla..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	NEMA 4X / IP66/68 Dual Compa..
Hazardous Class:	ATEX II 1/2G Ex ia IIC T6 Ga..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	4-20mA HART Level (0-100% / 0-6.0m)
Power Supply:	24V DC via 1756-IF16 Analog Channel
Field Cable Type:	1Px1.5mm ² Twisted Pair Shielded (Belde..
Destination JB:	JB-401 / JB-402
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	937THAITXPDC1 (IS Repeater Barrier)
PLC Module:	1756-IF16 (16-Point Differential AI)

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: ENDRESS+HAUSER

MODEL: Micropilot FMR67B (80 GHz High-Frequency Radar)

SERVICE: Cooker Slurry & Flash Tank Non-Contact Continuous Level

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Micropilot FMR67B Radar Transm E+H 80GHz w/ Drip-off PTFE Anten		1 EA
2	Tank Nozzle Flange DN80 Cl.150 SS316 Tank Flange Neck		1 EA
3	PTFE Envelope Gasket DN80 Cl.1 Chemical Resistant Gasket		1 EA
4	Cable Gland M20x1.5 Ex d/e IP6 Nickel-plated Brass Gland		1 EA
5	Air Purge Connection Kit 1/4" Purge Connection for Condensate		1 SET

ENGINEERING INSTALLATION NOTES

- Mount nozzle height must not exceed antenna length so horn protrudes clear into vessel interior.
- Distance to vessel wall should be min. 1/6 of vessel diameter to prevent false sidewall echoes.
- Activate Heartbeat Technology and mapping curves to mask agitator blade echoes during commissioning.

INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTAEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONRADAR LEVEL TRANSMITTER (80 GHz) HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDWG NO: KAL-JC-TYP-014
REV: 3.6 DATE: 2026-09-08DRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.PROJECT NO:
R5THS00138-JCSHEET NO:
16 / 25